$$d_{Lev}(a,b) = \begin{cases} |a|, & \text{if} \quad |b| = 0 \\ |b|, & \text{if} \quad |a| = 0 \\ d_{Lev}(a_{1\ldots n}, b_{1\ldots n}), & \text{if} \quad a_0 = b_0 \\ 1 + \min\left\{d_{Lev}(a_{1\ldots n}, b), d_{Lev}(a, b_{1\ldots n}), d_{Lev}(a_{1\ldots n}, b_{1\ldots n})\right\}, & \text{else} \end{cases}$$

## C. Model versions

For reproducibility, we include the model versions used to collect data in Table 1, alongside the providers.

| Model Name | Model Version | Provider |
|---|---|---|
| Claude-3.5-Sonnet | claude-3-5-sonnet-20241022 | Claude API |
| DeepSeek-V3 | DeepSeek-V3 | together.ai |
| GPT-4o | gpt-4o-2024-08-06 | Azure API |
| Llama-3.1-8b | Llama-3.1-8b-Instruct-Turbo-128K | together.ai |
| Llama-3.1-70b | Llama-3.1-70b-Instruct-Turbo | together.ai |
| Mixtral-8x22B | Mixtral-8x22B-Instruct-v0.1 | together.ai |

*Table 1.* Full model versions alongside the model provider used to access them.

## D. Probing Analysis

### D.1. Background on Language Models

When a sentence $s$ is fed into a language model, the text is first tokenized into a sequence of tokens, $x_1, ..., x_t \in V$, where $V$ is the vocabulary of the model. Then when generating the $x_{t+1}$ token, the model, $f : \mathcal{X} \to \mathcal{Y}$ maps the input sequence to a probability distribution $\mathcal{Y} \in \mathbb{R}^{|V|}$. This is done by first embedding each of the input tokens $x_i$ using a learned input embedding matrix $E$ that maps the vocabulary to a set of embeddings denoted by $h_i^{(0)}$. As the token is processed throughout the model, each token's latent layer is updated as follows:

$$h_i^{(j)} = h_i^{(j-1)} + g(h_1^{(j-1)}, ..., h_i^{(j-1)})$$

Here $g$ is a function that typically consists of an MLP and an attention layer, alongside some form of normalization. Using the final representation of the last token $h_x^{(L)}$, an unembedding matrix is applied, $W$, which converts the latent vector back into the vocabulary space. We use the internal latents of the language model to train the probe.

### D.2. Training and Evaluation

The input data to the probe is the residuals of the last token from the different layers. In our case, this maps to the ":" after "Rating:". It is important to note that instruct models require specific prompts using roles to stay faithful to the training process. To account for this, we put the original task into the user message, and put the "Rating:" as the first tokens of the assistant message. We then continued generating from these forced tokens.

The probe itself consisted of a single affine layer predicting one value (the similarity score). In other words, by learning the probe we were essentially learning a mapping from the hidden dimension of the model ($4096$) to a one-dimensional space.

To train the probe, we first extract residuals from 10,000 random pairs of numbers between 0-999 across all 32 layers of Llama-3.1-8b. We then train probes on all layers and with varying dataset sizes (see below for the analysis). The groundtruth data is calculated by measuring either the Log-Linear or Levenshtein distance between all pairs of numbers. Then the probe is trained to predict these scores. The probe itself is trained with Adam for 100 epochs.

In Table 2 we report the correlation between the probes and the groundtruth Levenshtein and log-linear judgments.

| | String Probe | Levenshtein | Int Probe | Log-Linear |
|---|---|---|---|---|
| String Probe | 1 | – | – | – |
| Levenshtein | 0.650*** | 1 | – | – |
| Int Probe | 0.667*** | 0.393*** | 1 | – |
| Log-Linear | 0.527*** | 0.266*** | 0.917*** | 1 |

*Table 2.* Correlation between the probes and the groundtruth number similarities based on Levenshtein and Log-Linear distance. All correlations have $p < 0.00001$.

## E. Linear Number Distance Control Analysis

As an additional control we reran our regression analysis with a linear $\ell_1$ number distance $|a - b|$ rather than the psychologically-motivated log linear one. The results are shown in Figure 6 and they yielded a worse fit relative to the Log-Linear case.

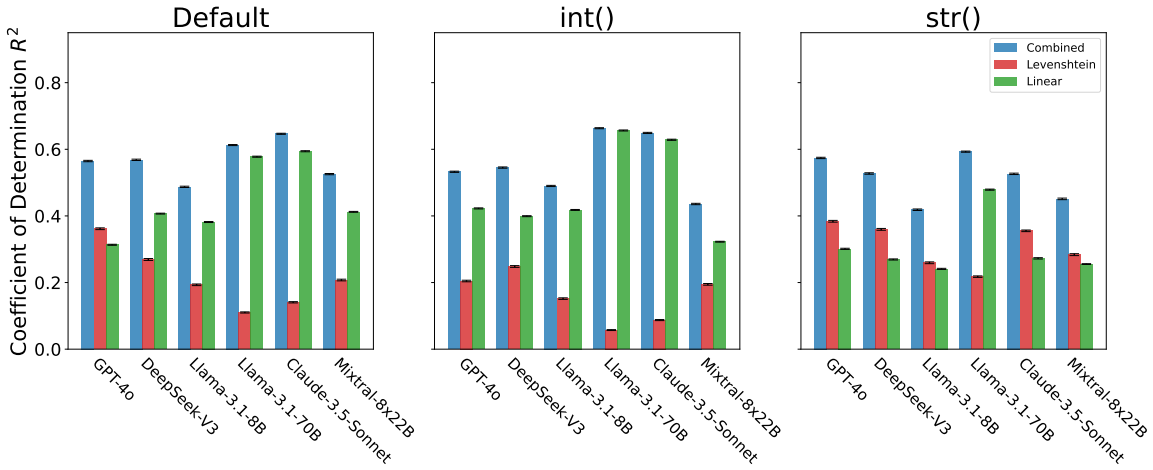


*Figure 6.* Coefficient of determination ($R^2$) for the different similarity matrices under the default (Figure 1), `int()`, and `str()` contexts for the combined and separate Levenshtein (string) and Linear (numerical) distance predictors (error bars indicate 95% confidence intervals; see Methodology).

## F. Extensions

### F.1. Number of Comparisons

Within the main paper, we restricted the similarity judgments to a maximum of $1000 \times 1000$. Here we extend this for GPT-4o to the $2000 \times 2000$, presented in Figure 8. We find that the similarity pattern continues after the number 1000, but begins to get broader, possibly because of tokenization effects.

### F.2. Other Qualifiers

Throughout the paper, we used the "similar" qualifier when asking the model to compare two numbers. This is because "similar" is inherently neutral and is better motivated from the psychological literature for exploring representations. As an exploratory analysis we recomputed the GPT-4o similarity matrix using an alternative "close" qualifier which has a more quantitative connotation (similar to the `int()` type). We provide a side-by-side comparison in Figure 8. As expected, we observe that the "close" case has an effect that more closely aligns with the `int()` context. Nevertheless, we still notice a pronounced set of string-induced diagonals, especially as the integers get larger.

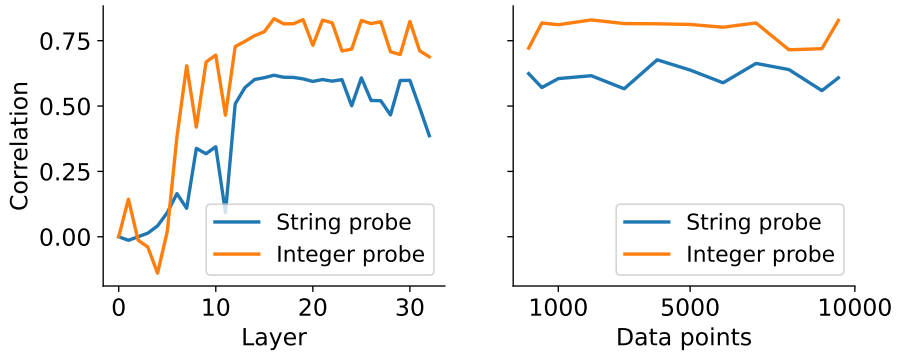


*Figure 7.* Ablations for the training the probe. The vertical axis shows the correlation between the groundtruth similarities and the probe similarities on the test set. The plot on the left features how the correlation varies with the layer the probe is trained on using the full 9,500 data points for training. The right plot illustrates how the number of data points affects the correlation for a probe on layer 25.

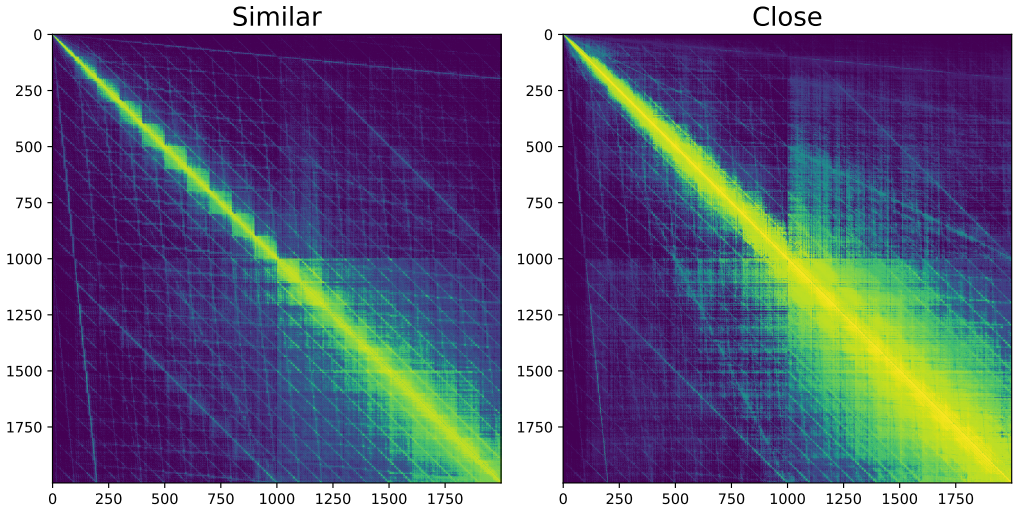


*Figure 8.* (left) Similarity judgments extended to the 2000 $\times$ 2000 setting. (right) Original prompt where "similar" is replaced with "closer" in the 2000 $\times$ 2000 setting. Both plots are for GPT-4o.

### F.3. Higher Temperature

In the final analysis, we re-run the temperature 0 setting in the "basic similarity prompt" context at a temperature of 0.7. Using the new similarity matrices we measure the correlation and mean absolute difference across runs. We find that across all models that correlation is above 0.787, and on average above 0.87. The model which had the lowest correlation was Mixtral-8x22B.